Exercise Sheet 20

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$ and

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

$$k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \quad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \quad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

On $\mathfrak{sl}(2,\mathbb{C})$ the Killing form is $\kappa(X,Y) = 4\,\mathrm{tr}(XY)$. The group G acts on $\mathbb{H} = \{z : \mathrm{Im}\, z > 0\}$ by

$$g \cdot z = (\alpha z + \beta)/(\gamma z + \delta),$$

where $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$.

Exercise 20.1 (The real form $\mathfrak{su}(1,1)$). Let $J = \mathrm{diag}(1,-1)$ and

$$\mathfrak{su}(1,1) = \{X \in \mathfrak{sl}(2,\mathbb{C}) : X^*J + JX = 0\}.$$

Put

$$U_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$

(a) Show that $\mathfrak{su}(1,1)$ consists of the matrices $\begin{pmatrix} ia & b \\ \bar{b} & -ia \end{pmatrix}$ with $a \in \mathbb{R}$, $b \in \mathbb{C}$. Show that it is a real form of $\mathfrak{sl}(2,\mathbb{C})$.

(b) Compute the Gram matrix of κ in the basis (U_1, U_2, U_3), and its signature.

(c) Compute cHc^{-1}, $c(E+F)c^{-1}$ and $c(E-F)c^{-1}$. Deduce that $X \mapsto cXc^{-1}$ is a Lie algebra isomorphism $\mathfrak{sl}(2,\mathbb{R}) \to \mathfrak{su}(1,1)$. Check that this agrees with (b).

Exercise 20.2 (Iwasawa decomposition of $\mathrm{SL}(n,\mathbb{R})$). Let $K = \mathrm{SO}(n)$. Let A be the positive diagonal matrices of determinant 1. Let N be the upper triangular matrices with all diagonal entries 1. Since n denotes the matrix size, write u for an element of N. Let $\mathfrak{a}$ be the real traceless diagonal matrices.

(a) Use Gram–Schmidt on the columns of $g \in \mathrm{SL}(n,\mathbb{R})$ to show $g = kau$ with $k \in K$, $a \in A$, $u \in N$.

(b) Show that $K \cap AN = \{I\}$. Deduce that the decomposition in (a) is unique.

(c) Take $H_0 = \mathrm{diag}(n-1, n-3, \ldots, 1-n) \in \mathfrak{a}$. Show that the positive restricted roots are $\lambda_{ij}(H) = h_i - h_j$ for $i < j$, and that $\mathfrak{n}$ is the space of strictly upper triangular matrices. Show that $A = \exp\mathfrak{a}$ and $N = \exp\mathfrak{n}$.

Exercise 20.3 (An explicit Iwasawa computation). Let $g = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

(a) Find φ, t, x with $g = k_\varphi a_t n_x$, and verify the product.
(b) Compute $g \cdot i$ directly.
(c) Find x, t and $k \in \mathrm{SO}(2)$ with $g = n_x a_t k$. Identify k.

Exercise 20.4 (Polar and KAK forms of a unipotent element). Let $g = n_2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Use the following fact. If Y is symmetric and traceless with $Y^2 = s^2 I$ and $s \geq 0$, then

$$\exp Y = \cosh s \, I + \frac{\sinh s}{s} Y \qquad \left(\tfrac{\sinh s}{s} := 1 \text{ at } s = 0\right).$$

(a) Compute $Q = g^T g$. Show that $P = (Q + I)/(2\sqrt{2})$ is symmetric positive definite with $\det P = 1$ and $P^2 = Q$.
(b) Compute $k = gP^{-1}$ and show $k = k_{-\pi/4}$.
(c) Find symmetric traceless X with $\exp X = P$.
(d) Show that $g = k_{\pi/8} \, a_t \, k_{-3\pi/8}$ with $t = \log(1 + \sqrt{2})$.

Exercise 20.5 (K-orbits in the upper half-plane). Let $K = \mathrm{SO}(2)$ act on $\mathbb{H}$ as above.

(a) For $g \in G$ and $z, w \in \mathbb{H}$, show that

$$g \cdot z - g \cdot w = \frac{z - w}{(\gamma z + \delta)(\gamma w + \delta)}.$$

(b) Deduce that $\Phi(z) = |z - i|^2 / \operatorname{Im} z$ is K-invariant.
(c) For $t \geq 0$, show that the orbit $K \cdot (ie^{2t})$ is the circle with centre $i \cosh 2t$ and radius $\sinh 2t$. You may use the lecture's fact that every $z \in \mathbb{H}$ equals $k \cdot (ie^{2s})$ for some $k \in K$ and $s \geq 0$.
(d) Show that the set of $g \in G$ acting trivially on $\mathbb{H}$ is $\{\pm I\}$.

Exercise 20.6 (Optional extension: the affine group and the invariant area). Use the affine group of Lecture 16. Its elements are the pairs (y, b) with $y > 0$ and $b \in \mathbb{R}$, and its product is

$$(y, b)(y', b') = (yy', b + yb').$$

Its left Haar measure is $dy \, db / y^2$.

(a) Show that $\Psi(n_x a_t) = (e^{2t}, x)$ is an isomorphism from $NA = \{n_x a_t\}$ onto the affine group.
(b) Show that $n_x a_t \mapsto n_x a_t \cdot i$ carries left translation on NA to the action on $\mathbb{H}$. Deduce that $dx \, dy / y^2$ on $\mathbb{H}$ is NA-invariant.
(c) Show directly that $dx \, dy / y^2$ is invariant under all of G. Use the fact that a holomorphic map $w(z)$ has real Jacobian determinant $|w'(z)|^2$.